Sep 2 CDF X

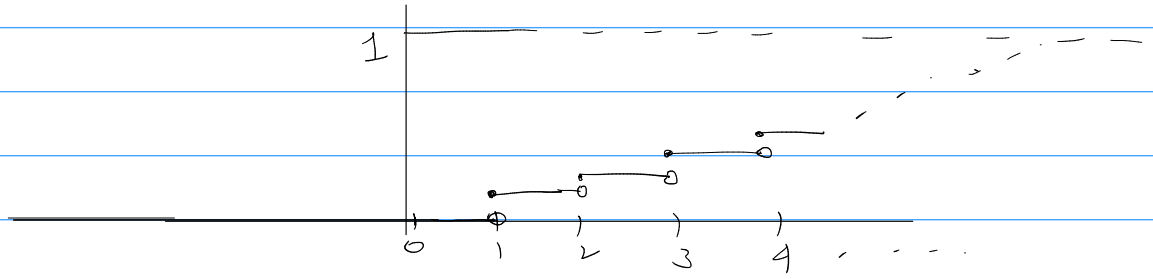
$$F_X(x) = P(X \le x)$$

① $X \sim \text{Geometric}(p)$ $\quad p_X(k) = (1-p)^{k-1} p$ $\quad q = 1-p$

$k \in 1, 2, 3, \ldots$

$$F_X(x) = \sum_{k=1}^{\lfloor x \rfloor} p_X(k) = p \cdot \frac{1-(1-p)^{\lfloor x \rfloor}}{1-(1-p)} = 1 - q^{\lfloor x \rfloor}$$

$\lfloor x \rfloor$: largest integer $\le x$



② $X \sim \text{Poisson}(\lambda)$

$$F_X(x) = \sum_{k=0}^{\lfloor x \rfloor} p_X(k) = e^{-\lambda} \sum_{k=0}^{\lfloor x \rfloor} \frac{\lambda^k}{k!}$$

③ $X \sim \text{Binomial}(n, p)$

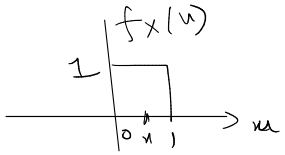
$$F_X(x) = \sum_{k=0}^{\lfloor x \rfloor} p_X(k) = \begin{cases} 0, & x < 0 \\ \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k (1-p)^{n-k}, & 0 \le x \le n \\ 1, & x > n \end{cases}$$

Properties of a CDF: $F_X : \mathbb{R} \to \mathbb{R}$

Valid CDF:

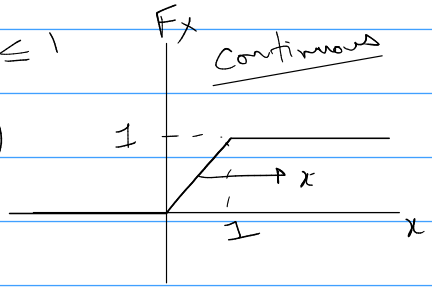
- $F_X(-\infty) = 0$, $F_X(\infty) = 1$
- F_X is non-decreasing
- F_X is continuous from the right

Ex: ① $X \sim \text{Uniform}[0,1]$ PDF: f_X



$$F_X(x) = \begin{cases} 0, & x < 0 \\ P(0 \le X \le x) = x, & 0 \le x \le 1 \\ 1 & x > 1 \end{cases}$$

$$\int_0^x f_X(u)\,du = x$$

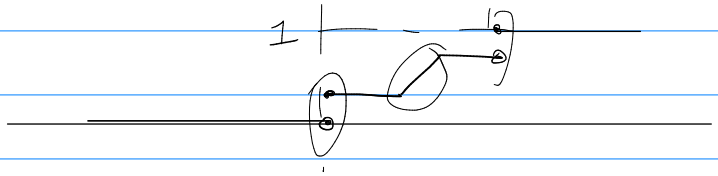


Valid CDFs

→ Discrete RVs → jumps at a discrete set of points
no increase other than at jumps

→ Continuous RVs → F_X is continuous (no jumps)

→ Mixed RVs → both occur



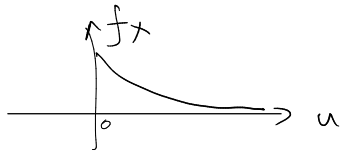
Defn of PDF: If there is an f_X s.t. the CDF $F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$, then

f_X is the PDF of X.

→ F_X: differentiable

$$f_X = F_X'$$

Ex: ① $X \sim$ PDF e^{-x}, $x > 0$.

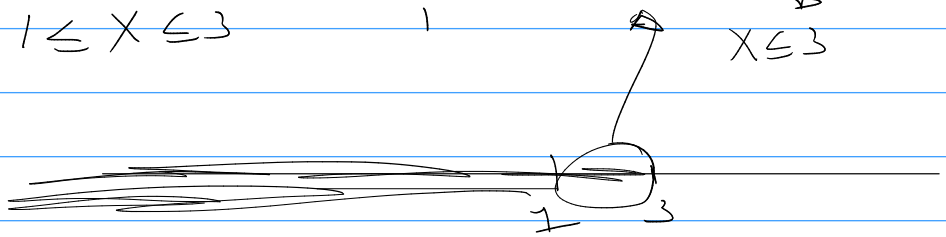


$$F_X(x) = P(X \le x) = \begin{cases} 0 & x < 0 \\ \int_0^x e^{-u}\,du & x \ge 0 \\ = 1 - e^{-x} & \end{cases}$$

derivative e^{-x}

$$P(X \in [1,3]) = \int_1^3 f_X(x)\,dx = F_X(3) - F_X(1)$$

$1 \le X \le 3$; $X \le 3$; $X \le 1$



CDF: computing probabilities of intervals

$$F_X(x) = P(X \le x)$$

interval $(a, b]$

$$F_X(b) - F_X(a) = P(X \le b) - P(X \le a)$$

matters if discrete case

$$= P(a < X \le b)$$

$(a, b]$

$$F_X(b) - F_X(a^-) = P(a \le X \le b)$$

$[a, b]$

Conditional probs: $b > a$

$$P(X > b \mid X > a) = \frac{P(X > b)}{P(X > a)}$$

Event $\{\omega \in S : X(\omega) > b\}$

$$= \frac{1 - F_X(b)}{1 - F_X(a)}$$

Bivariate distributions

(S, event, P)

Discrete: X, Y : two RVs defined on same probability space.

$X \in \mathcal{X}$, $Y \in \mathcal{Y}$

Marginal PMF for X: $p_X(x) = P(X = x)$

Marginal PMF for Y: $p_Y(y) = P(Y = y)$

$\{\omega \in S : X(\omega) = x\}$

$\{\omega \in S : Y(\omega) = y\}$

Joint PMF of X, Y: $p_{XY}(x, y) = P(X = x \cap Y = y)$

$\{\omega \in S : X(\omega) = x \text{ and } Y(\omega) = y\}$

$= P(X = x, Y = y)$